



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject : **SECP2613 System Analysis and Design (WBL)**
Task : **Assignment 1**
Title : **Group Assignment 1 (Project Planning 5%)**
Due : **25/4/2024**
Submission : **Submit softcopy (in .pdf format) via e-learning**

INSTRUCTIONS:

1. 3 - 4 students per group
2. All answers should be uploaded via link specified in e-learning.
3. Any plagiarism – copying directly from other groups will result in both groups will be penalized by getting an 'F' grade.

QUESTION 1 (10 MARKS)

"We could really make some changes. Shake up some people. Let them know we're with it. Technologically, I mean," said Ahmad, vice president for ALLFOODS, a wholesale dairy products distributor. "That old system should be overhauled. I think we should just tell the staff that it's time to change."

"Yes, but what would we actually be improving?" Abu, assistant to the vice president, asks. "I mean, there aren't any substantial problems with the system input or output that I can see."

Ahmad snaps, "Abu, you're purposely not seeing my point. People out there see us as a stodgy firm. A new computer system could help change that. Change the look of our invoices. Send a report to the food store owners. Get some people excited about us as leaders in wholesale food distributing and computers."

"Well, from what I've seen over the years," Abu replies evenly, "a new system is very disruptive, even when the business really needs it. People dislike change, and if the system is performing the way it should, maybe there are other things we could do to update our image that wouldn't drive everyone nuts in the process. Besides, you're talking big bucks for a new gimmick."

Ahmad says, "I don't think just tossing it around here between the two of us is going to solve anything. Check on it and get back to me. Wouldn't it be wonderful?"

A week later Abu enters Ahmad's office with several pages of interview notes in hand. "I've talked with most of the people who have extensive contact with the system. They're happy, Ahmad. And they're not just talking through their hats. They know what they're doing."

"I'm sure the managers would like to have a newer system than the guys at Quality Foods," Ahmad replies. "Did you talk to them?" Abu says, "Yes. They're satisfied."

"And how about the people in systems? Did they say the technology to update our system is out there?" Ahmad inquires insistently. "Yes. It can be done. That doesn't mean it should be," Abu says firmly.

As the systems analyst for ALLFOODS,

- a. how would you assess the feasibility of the systems project Ahmad is proposing?
(2 Marks)
- b. Based on what Abu has said about the managers, users, and systems people, what seems to be the operational feasibility of the proposed project?
(2 Marks)
- c. What about the economic feasibility?
(2 Marks)
- d. What about the technological feasibility?
(2 Marks)
- e. Based on what Ahmad and Abu have discussed, would you recommend that a full-blown systems study be done? Discuss your answer in a paragraph.
(2 marks)

QUESTION 2 (10 MARKS)

Think of all the tasks that you perform when you want to propose a new computer system for student registration. Include any research, decisions, or financial issues that related to the development of new computer system. Create a Work base Schedule (WBS) that shows all the tasks, their estimated duration, and any predecessor tasks.

QUESTION 3 (10 MARKS)

Many of today's projects involve team members scattered across different time zones and in different physical locations. Moreover, the projects may have adopted an agile methodology, which reduces cycle time dramatically. Write a brief report that summarizes some of the key differences a manager would face managing this type of project, as opposed to a traditional project.

QUESTION 4 (10 MARKS)

SKORBISTARI leading preschool was established with the dream of bringing quality of preschools to greater high based on Islamic principles, English emphasis, and cheerful well-designed facilities. The company grows to over 53 centres all over the country. Recently, the top management has decided to consider an information system to help their growing business. You have been hired as the system analyst to handle the project. Before continue with the development of the project, you need to assess the economic feasibility from the budget that the company plans to invest. Table 1 shows the information given by the company.

Table 1: Estimated cost and expected benefits for SKORBISTARI

Development Costs	
Hardware	RM 50,000
Software	RM 15,000
Training	RM 30,000
Consulting	RM 80,000
Data Conversion	RM 40,000
Production Costs	
Supplies	RM 15,000 per year
IS Salaries	RM 75,000 in year 1; 10% annual increase
Upgrades	RM 10,000 per year over 3 years
Benefits	
Improve Customer Service	RM 150,000 in year 1; 20% increase annually
Increase productivity	RM 150,000 per year; 25% increase annually
Assumptions	
Discount rate	35 %

- (a) Calculate the cost-benefit estimation using the Present Value (PV) analysis (for 3 years) to assess the economic feasibility for the SKORBESTARI project proposal in the worksheet.

(9 marks)

- (b) What is the profitability index (PI) value for this PV analysis? What is your recommendation based on the PI value and justify it.

(1 marks)

COST	YEAR 0	YEAR 1	YEAR 2	YEAR 3
DEVELOPMENT COSTS				
Hardware	50000			
Software	15000			
Training	30000			
Consulting	80000			
Data Conversion	40000			
TOTAL COST	215000			
Total				
PRODUCTION COST				
Supplies		15000	15000	15000
IS Salaries		75000	82500	90750
Upgrades		10000	10000	10000
ANNUAL PRODUCTION COST		100000	107500	115750
$\text{Payment} \times (1/(1+c)^n)$ (PRESENT VALUE)		74074	58984	47046
ACCUMULATED COST		289074	348058	395104
BENEFIT				
Improve Customer Service		150000	180000	216000
Increase productivity		150000	187500	234375
ANNUAL BENEFIT		300000	367500	450375
(PRESENT VALUE)		222222	201646	183051
ACCUMULATED BENEFIT		222222	423868	606919
Benefit - Cost		(66852)	75810	211815
GAIN OR (LOSS)				
PROFITABILITY INDEX	0.985			
JUSTIFICATION	Profitability index = 0.985, showing that it is not a good investment because its index is less than one.			

QUESTION 5 (10 MARKS)

A project has been defined to contain the following list of activities along with their required times of completion.

Activity No	Immediate Activity	Time (weeks)	Predecessors
1	Collect requirements ✓	3	
2	Analyze processes ✓	2	1
3	Analyze data ✓	2	2
4	Design Process ✓	6	2
5	Design data ✓	3	3
6	Design Screens ✓	2	3, 4
7	Design reports ✓	4	4, 5
8	Program ✓	5	6, 7
9	Test and document ✓	7	7
10	Install ✓	2	8, 9

a. Draw a PERT Diagram for the activities.

(3 Marks)

b. Lists all the paths.

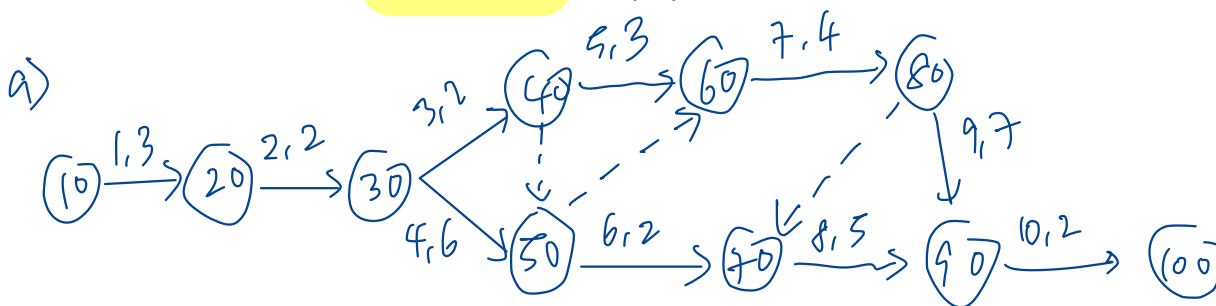
(2 Marks)

c. Calculate and identify the critical path.

(2 Marks)

d. Construct a Gantt chart for the project defined above.

(3 Marks)



Path 1: 1 - 2 - 3 - 5 - 7 - 9 - 10 = 23

Path 2: 1 - 2 - 4 - 6 - 8 - 10 = 20

Path 3: 1 - 2 - 3 - 6 - 8 - 10 = 16

Path 4: 1 - 2 - 4 - 7 - 9 - 10 = 24

Path 5: 1 - 2 - 4 - 7 - 8 - 10 = 22

Path 6: 1 - 2 - 3 - 5 - 7 - 8 - 10 = 21

Critical Path: Path 4: 1 - 2 - 4 - 7 - 9 - 10

QUESTION 1

- a) I will assess the feasibility of the system project by looking at technical, economic, and operational aspects. This necessitated an examination of the technology available to suit users' needs, the expense of system research, and whether the system would be used.
- b) Based on Abu's statements, the operational feasibility suggests that nothing should change. This is because relevant stakeholders are happy and comfortable with the current system, and Abu believes there are no significant flaws with it.
- c) Economic feasibility entails calculating the cost-benefit analysis (CBA) for implementing a new system and determining if time and funds are available to create it. Abu stated that the shift required a large expenditure. Furthermore, while the current system is still running properly, the potential benefit of the new system is negligible.
- d) Technological feasibility determines whether the existing technical resources are adequate for the new system. Abu has spoken with the development team, and they all agree that the technology to update the system is available. Therefore, this project is realizable.
- e) Based on what Ahmad and Abu have discussed, I advocate not proposing a full-blown system now. This is due to high stakeholder satisfaction with the current system, which may prevent the new system from becoming operationally feasible, as well as being pricey. I agree with Abu that they can use various techniques to improve the company's image.

QUESTION 2

Work Breakdown Structure (WBS) for Student Registration Computer System

Activity	Duration (Weeks)	Predecessor
1.0 Planning	4	None
1.1 Define problems and determine goals for new system		
1.1.1 Interview and questionnaire		
1.2 Form project management team		
1.3 Conduct feasibility study		
1.3.1 Defining objectives		
1.3.2 Determining resources		
1.3.2.1 Operational feasibility		
1.3.2.2 Technical feasibility		
1.3.2.3 Economic feasibility		
1.3.2.3.1 Perform Cost-Benefit Analysis		
1.4 Project scheduling		
1.4.1 Gantt Chart		
2.0 Analysis	2	Planning
2.1 Determine end users needs and clarify specific details required for new system		
2.1.1 Interview and questionnaire		
3.0 Design	6	Analysis
3.1 Define overall system architecture		
3.2 Design algorithms and data structure		
3.3 Create data flow diagrams		
3.4 Prepare and present system proposal		
4.0 Development	8	Design
4.1 Develop software		
4.2 Create documentation		
4.2.1 Create procedure manuals		
4.2.2 Create Websites with Frequently Asked Questions		

5.0 Testing	4	Development
5.1 Security testing		
5.2 Performance testing		
5.3 Usability testing		
5.4 Compatibility testing		
6.0 Implementation	4	Testing
6.1 Do user training		
6.2 Install system		
6.3 Converting from old system to new system		
6.4 Data migration		

QUESTION 3

Report: Main Differences in Managing Agile Projects vs Traditional Projects

Introduction

As modern project management continues to develop, distributed teams and agile methodologies become more prominent. This report will describe the fundamental differences in managing agile projects with distributed teams versus traditional projects with co-located teams.

Key Differences

1. Communication Challenges

Agile projects rely heavily on frequent and open communication. Due to limited face-to-face encounters, time zone differences, and geographical dispersion, agile projects using distributed teams have considerable communication issues. Team members must rely extensively on virtual communication channels such as emails, video calls, and messaging apps. On the other hand, traditional projects gain from co-located teams' easier and faster communication, which speeds up decision-making and problem-solving.

2. Collaboration Dynamics

Agile distributed teams rely on online platforms and technologies for virtual collaboration to exchange work, give feedback, and manage tasks. This reliance on technology adds a level of complication that traditional initiatives may not encounter to the same degree. In traditional projects, in-person collaboration encourages direct and spontaneous interactions among team members, which can boost productivity and teamwork.

3. Cultural Sensitivity

People in distributed agile teams frequently come from different cultural backgrounds, thus understanding cultural norms and variances is important. To create a cohesive work atmosphere, managers need to encourage cultural awareness, inclusivity, and successful cross-cultural communication. Traditional projects may have more culturally homogeneous teams, simplifying the management of cultural diversity.

4. Time Zone Coordination

One of the most important aspects of managing distributed agile teams is managing time zone differences. Careful coordination and flexibility are needed for scheduling meetings, coordinating project updates, and guaranteeing ongoing communication across time zones. On the other hand, traditional projects with co-located teams encounter fewer issues due to time zone variations, allowing for more efficient workflow management and communication.

5. Autonomy and Empowerment

Agile methodologies foster autonomy and empowerment among team members, enabling them to make decisions and accept responsibility for their work. Because there is less direct supervision, distributed agile teams frequently operate with more autonomy, which makes it necessary to establish trust and communicate expectations clearly. In traditional projects, team members may receive more direct supervision and coaching from supervisors.

Conclusion

Managing agile projects with distributed teams poses unique problems when compared to traditional projects with co-located teams. A combination of cultural sensitivity, adaptation, technological proficiency, communication skills, and risk management techniques suited to the unique dynamics of each project environment are needed for effective management in both situations.

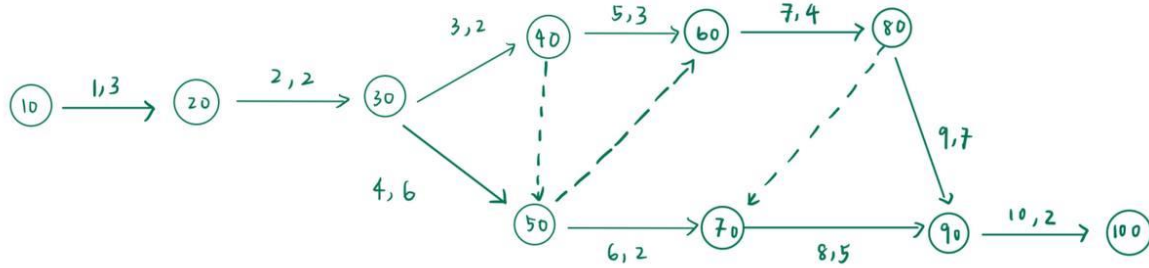
QUESTION 4

a.

	A	B	C	D	E
1	Costs	Year 0	Year 1	Year 2	Year 3
2					
3	Development Costs				
4	<i>Hardware</i>	50000			
5	<i>Software</i>	15000			
6	<i>Training</i>	30000			
7	<i>Consulting</i>	80000			
8	<i>Data Conversion</i>	40000			
9	Total	215000			
10					
11	Production Costs				
12	<i>Supplies</i>		15000	15000	15000
13	<i>IS Salaries</i>		75000	82500	90750
14	<i>Upgrades</i>		10000	10000	10000
15	Annual Prod.Costs		100000	107500	115750
16	(Present Value)		74074	58984	47046
17	Accumulated Costs		289074	348058	395104
18					
19	Benefits				
20	<i>Improved Customer Service</i>		150000	180000	216000
21	<i>Increase Productivity</i>		150000	187500	234375
22	Annual Benefits		300000	367500	450375
23	(Present Value)		222222	201646	183051
24	Accumulated Benefits		222222	423868	606919
25					
26	Gain or Loss		-66852	75810	211815
27					
28	Profitability Index	0,985			
29					

b. Profitability index = 0.985, showing that it is not a good investment because its index is less than one.

5a)



b)

Path 1: 1-2-3-6-8-10

Path 2: 1-2-3-5-7-9-10

Path 3: 1-2-3-5-7-8-10

Path 4: 1-2-4-7-8-10

Path 5: 1-2-4-6-8-10

Path 6: 1-2-4-7-9-10

c) Critical path:

Path 1: 1-2-3-6-8-10=16

Path 2: 1-2-3-5-7-9-10=23

Path 3: 1-2-3-5-7-8-10=21

Path 4: 1-2-4-7-8-10=22

Path 5: 1-2-4-6-8-10=20

Path 6: 1-2-4-7-9-10=24

Therefore, critical path which is longest path is path 6: 1-2-4-7-9-10

d)

